Introduction to Generative AI

Generative AI is a branch of artificial intelligence that focuses on creating new content such as text, images, audio, and code. It has gained massive popularity with the rise of large language models.

Key technologies include transformers, neural networks, and large-scale datasets. These models are trained on billions of parameters.

Real-world applications include chatbots, content generation, coding assistants, and automation tools across industries.

Core Foundations

To master Generative AI, a strong foundation in Python is essential. Understanding data structures, APIs, and libraries is crucial.

Basic knowledge of machine learning concepts like supervised and unsupervised learning helps in understanding AI pipelines.

Understanding how APIs work is important because most LLMs are accessed via APIs.

Understanding LLMs

Large Language Models (LLMs) are trained on vast amounts of text data and can generate human-like responses.

Concepts such as tokens, embeddings, and attention mechanisms are fundamental to understanding how LLMs work.

Popular LLMs include GPT models, which are widely used for conversational AI and automation.

Prompt Engineering

Prompt engineering is the process of designing inputs to get desired outputs from LLMs.

Techniques include zero-shot prompting, few-shot prompting, and chain-of-thought reasoning.

Well-structured prompts significantly improve model accuracy and reliability.

Embeddings and Vector Databases

Embeddings convert text into numerical vectors that capture semantic meaning.

Vector databases like Chroma and FAISS store embeddings and allow similarity-based retrieval.

This is essential for building intelligent search systems and RAG applications.

Retrieval Augmented Generation

RAG combines retrieval and generation to provide accurate and context-aware responses.

It reduces hallucination by grounding responses in real data.

Pipeline includes loading data, splitting, embedding, storing, retrieving, and generating answers.

LangChain Fundamentals

LangChain is a framework that simplifies building LLM-powered applications.

It provides tools for document loading, splitting, embeddings, and chaining operations.

It is widely used for building RAG systems and AI-powered workflows.

Building RAG Applications

RAG applications allow users to query documents such as PDFs and get precise answers.

Examples include chatbots for customer support and knowledge assistants.

These systems rely on vector databases and retrievers.

LangGraph for Agents

LangGraph enables building stateful workflows for AI agents.

It allows multi-step reasoning and decision-making processes.

It is useful for building autonomous AI systems.

LangSmith for Monitoring

LangSmith helps in debugging and monitoring AI applications.

It provides tracing, logging, and evaluation tools.

It ensures reliability and performance of AI systems.

Building AI Agents

AI agents can use tools, APIs, and external data sources to perform tasks.

They can be designed for automation, research, and decision-making.

Multi-agent systems allow collaboration between multiple AI agents.

Production and Scaling

Deploying AI applications requires scalability and performance optimization.

Techniques include caching, batching, and efficient API usage.

Monitoring and evaluation are critical for maintaining quality in production systems.